

# K-Dash Pro - Product Requirements

## 1. Project Overview

K-Dash Pro is an enterprise-grade analytics dashboard designed to monitor, analyze, and manage business operations through a centralized interface. The system aggregates data from multiple business domains—including sales, inventory, customers, and system health—and presents it in an interactive, real-time dashboard.

The primary objective is to help decision-makers quickly understand business performance, identify operational issues, and make data-driven decisions.

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## 2. Business Goals

The application is designed to achieve the following objectives:

- Provide real-time visibility into business operations.
  - Enable fast decision-making through data visualization.
  - Reduce the time required to identify operational issues.
  - Improve productivity with powerful filtering and reporting tools.
  - Offer a scalable architecture suitable for enterprise applications.
  - Demonstrate modern Angular development practices and enterprise UI patterns.
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## 3. Target Users

### Administrator

Responsible for managing the entire system.

Capabilities:

- Manage users and roles
  - Configure application settings
  - Access all reports
  - Create, update and delete data
  - View audit logs
  - Manage permissions
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## Manager

Responsible for monitoring business performance.

Capabilities:

- View dashboards
- Analyze reports
- Filter and export data
- Manage inventory and orders
- Receive operational alerts

Restrictions:

- Cannot delete critical data
  - Cannot manage users
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## Viewer

Responsible for monitoring information only.

Capabilities:

- Read-only access
- View dashboards
- View reports
- View notifications

Restrictions:

- No editing
  - No deleting
  - No administration
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# 4. Functional Requirements

## Dashboard

The dashboard shall provide:

- KPI Summary Cards
- Revenue Overview
- Sales Trends
- Inventory Status

- Recent Orders
  - Recent Transactions
  - Notification Panel
  - System Health Status
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## Analytics

The analytics module shall include:

- Line Charts
  - Area Charts
  - Bar Charts
  - Pie Charts
  - Trend Analysis
  - Date Range Filtering
  - Drill-down Navigation
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## Orders

The system shall allow users to:

- View orders
  - Search orders
  - Filter orders
  - Sort orders
  - Export orders
  - View order details
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## Inventory

The inventory module shall provide:

- Product listing
  - Inventory status
  - Low stock alerts
  - Category filtering
  - Supplier information
  - Bulk actions
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## Customers

The customer module shall support:

- Customer list
  - Customer profile
  - Purchase history
  - Customer statistics
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## User Management

The system shall support Role-Based Access Control (RBAC).

Roles:

- Administrator
- Manager
- Viewer

Permissions shall control:

- Page access
  - Menu visibility
  - UI actions
  - API operations
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## Notifications

The application shall display:

- Low inventory alerts
  - Failed payment alerts
  - System notifications
  - Success messages
  - Warning messages
  - Error messages
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## Reports

The reporting module shall support:

- Sales reports

- Inventory reports
- Customer reports
- Revenue reports

Reports can be exported as:

- CSV
  - Excel
  - PDF
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## 5. Non-Functional Requirements

The application shall satisfy the following quality attributes:

### Performance

- Fast initial loading
- Lazy-loaded modules
- Virtual scrolling for large datasets
- Optimized rendering using OnPush Change Detection

### Scalability

- Modular architecture
- Feature-based organization
- Reusable components
- Shared design system

### Security

- JWT authentication
- Route Guards
- Permission Guards
- HTTP Interceptors
- Role-Based Access Control

### Reliability

- Global error handling
- Retry mechanisms
- Offline detection
- Graceful error states

### Accessibility

- Keyboard navigation
- Focus management

- Screen reader compatibility
- WCAG-friendly color contrast

## **Responsiveness**

The application shall provide an optimized experience across:

- Desktop
  - Tablet
  - Mobile devices
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# **6. User Experience Requirements**

The user interface shall provide:

- Clean and minimal design
  - Collapsible sidebar
  - Sticky top navigation
  - Dark mode
  - Light mode
  - Responsive layout
  - Consistent spacing
  - Rounded corners
  - Soft shadows
  - Interactive feedback
  - Smooth animations
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# **7. Technical Requirements**

The application shall be implemented using:

- Angular Standalone Components
- Angular Signals (where appropriate)
- NgRx Store
- RxJS
- Angular Material
- CDK Drag & Drop
- ApexCharts or ECharts
- MSW or JSON Server
- Jasmine & Karma
- ESLint
- Prettier
- GitHub Actions

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## 8. Success Criteria

The project will be considered successful if it:

- Demonstrates enterprise-level Angular architecture.
- Provides a responsive and accessible user experience.
- Supports real-time dashboard interactions.
- Implements reusable and maintainable UI components.
- Uses scalable state management.
- Includes automated testing.
- Simulates realistic backend behavior through mock APIs.
- Is production-ready from an architectural perspective.